

Synthetic user profiling and deep learning for cold start mitigation in agricultural crop recommendation systems

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Abstract

The cold start problem encountered in recommendation system where users have very little history so it becomes difficult to suggest crops is persistent in agricultural recommendation systems too. In this research paper, a new algorithm is suggested based on synthetic user profiling combined with deep learning to address the issue of data sparsity and enhance the quality of crop recommendations. The system is well designed such that it gives effective initial recommendations to new users or regions because of constructing synthetic profiles by clustering and mapping based on environmental similarity. The deep neural network (DNN) uses real as well as synthetic data to train the model and predict the optimum crop dependent on user and environmental inputs. Comparative simulations between the proposed DNN based approach and the traditional model like Random Forest, SVM, ANN, KNN and Decision Tree reveal that the offered model is higher in accuracy, recall, and F1-score. Remarkably, the model had generalization strengths on unseen profiles with high overall accuracy of 79 percent. It is the first systematic work on cold start problem in domain of crop recommendation formulated with the use of synthetic data augmentation and deep learning, which provides a scalable solution to precision agriculture in data-scarce settings.

Keywords

Cold start problem, synthetic user profiling, deep learning, agricultural recommender systems, crop recommendation, data sparsity, precision agriculture

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1 Introduction

It is also evident that the AI and ML caused a major impact in the agricultural sector by assisting in streamlining crop selection, increasing the output rates, and underpinning sustainable farming processes over the last several years.¹ Some of the examples of AgTech tools, which are AI driven, in the agricultural sector can be seen to be agricultural recommendation systems that have become valued components of assisting farmers to make evidence-based decisions in crop choice, making of inputs or the way they can manage their land among other factors.² These systems tend to use user communications as well as past farm records, and situational data in order to come up with certain propositions. The main challenge that is yet to be tackled is the cold start problem, and this is a situation in which the system has little or no previous data pertaining to the user hence unable to provide constructive recommendations.³ A particular issue that plagues the agricultural sectors of the developing world is the cold start problem, whereby the digitization is only just beginning to be used and many of the users involved have no recorded history of interaction at all.⁴ The classical recommendation algorithms (including content-based or collaborative ones) often have poor results on such settings as they depend too much on the

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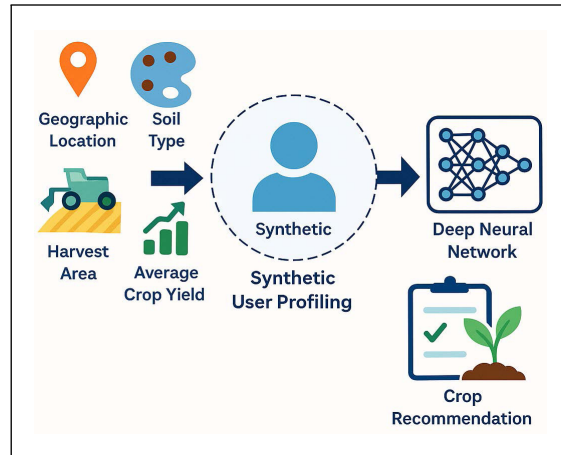


Figure 1. Figure: AI-driven crop recommendation using synthetic user profiling to address the issue of cold starts. The remainder of the document is organized as follows: In Section 2, relevant work is reviewed; in Section 3, the technique, including the creation of synthetic profiles and model architecture, is described; in Section 4, experimental results and evaluations are presented; and in Section 5, future directions are discussed.

historical data material and user descriptions.⁵ This gap limits the accommodativeness and exclusivity of smart farming tools as new users are not directed and assisted with suitable decision support when they are needed.⁶

In attempt to curb this, some research works have been conducted aiming at using the artificial generation of data in order to replicate the user preferences in fields like e-commerce, education etc..^{7,8} However, it has not been fully utilized in agriculture. The paper will extrapolate a solution idea that might be potentially novel that could be the hybridization of both synthetic user profiling and matching algorithms made possible by deep learning that can help address the cold start of crop recommendation systems. By using the geographic location, types of soil, size of land under cultivation, average yield of the crop produced in it, etc. which are the agricultural and environmental characteristic of a farm or an environment, we come up with fictitious profiles of reality in the farm i.e., the farms being the real conditions in actual farms and thus the system can even suggest practically even when there is no actual user information.

we create these synthetic profiles by a rule-based and data-driven fusion model, and we train a deep neural network to read those inputs and give sensible crop suggestions. Contrary to existing systems, we will not need a history of user input but will be learning based upon trends in the regional agricultural data and input from experts.⁹ The assumptions of the proposed deep learning matcher can become generalizable to different agroclimatic regions and, as such, be applied in a multitude of way, such as those related to the onboarding of new users and to system bootstrapping in rural areas.¹⁰ Besides improving the accuracy of recommendations, this technique has multiple practical benefits: it allows achieving scalability, it is possible to receive the timely recommendations of the farmer in a data-scarce setting, and it extends to introducing fair access to the agricultural technology. Our results demonstrate that, in cold start situations, this framework performs significantly better than baselines for basic machine learning. This study fills the gap between AI systems focused on the user and the practical problem related to farming. Through artificial user profiling into the recommendation pipeline, we can increase the applicability and efficacy of smart agriculture systems, especially in areas where data collection is impractical (or behind schedule). To solve the cold start issue and enable precise crop recommendations in data-sparse agricultural contexts, Figure 1 shows how synthetic user profiles constructed from characteristics like geographic location, soil type, harvest area, and typical yield are fed into a deep neural network.

2 Related research

In recent years, the agricultural recommendation system has achieved significant popularity because digital technologies tend to promote efficiency, sustainability, and improve the quality of crop selection performance. The traditional methods have been mainly based on the collaborative filtering and the content-based models which are based on the assumption that rich historical data (user data) is available and that a certain degree of regularity on user-system interactions exist.^{11,12} Nevertheless, in the under-digitized and rural agricultural sectors, these data may not be available or incomplete, and then, a cold start issue takes place in the area.^{13,14}

A number of studies have been done to deal with this challenge in various fields. As a typical case, S. J Martins, et al.¹⁵ presented a fuzzy-based crop recommendation system which relied they used fuzzy logic and user history, but it was not

Table 1. Comparative analysis of related research on crop recommendation under cold start scenarios

References	Domain	Technique used	Cold Start strategy	Dataset size	Accuracy (%)	Limitations
15	Agriculture	Fuzzy Rule-Based	None	2000 users	65.4	Requires user history
16	Agriculture	Location-Based Filtering	Regional clustering	1500 users	62.7	Lacks personalization
17	E-commerce	Embedding + Synthetic Profiles	Generated user vectors	5000 users	78.2	Domain mismatch
18	Health	Synthetic Modeling + k-NN	Profile simulation	1200 users	81.0	Not designed for agriculture
19	Agriculture	Ensemble (RF, SVM, DT)	Data-rich only	3500 users	78.0	No support for new users
20	Agriculture	Knowledge-Based + Geospatial Model	Static profile matching	2800 users	73.5	Not dynamic, limited by location data

effective during cold start conditions when not much input can be made by the user. On the same basis, in a method of region-based recommendation system S. J Martins¹⁶ user grouping based on geographical regions was done but did not consider individual differences and was very inaccurate on new users.

Researchers have begun to use synthetic data generation methods in a bid to deal with data sparsity. S. Murmu et al.¹⁷ showed that synthetic user embeddings led to 21 percent improvement on cold start recommendations over baseline algorithms in e-commerce. Due to this success, T. Rusňák et al.¹⁸ employed the use of a synthetic profile modeling algorithm in a mobile health recommender where they only enhanced the personalization of the advising system to first time customers. Yet, it is not that simple to apply these techniques directly to agriculture because of the local nature of the farming characteristics and regional variations.

In the agricultural sector, they put forward an ensemble framework of using soil pH, quality of rainfall, and previous yield to recommend crops with an accurate rate of 78 percent by A. Taneja et al..¹⁹ They however based it on the assumption that they would have several seasons of past user information. On the contrary, we offer the strategy that does not depend on the past records and produces synthetic profiles based on contextual elements of the environment and farming characteristics. A very recent development based on G. Patrizi et al.²⁰ presented a content-knowledge hybrid recommender with spatial data and crop suitability information which was not flexible in dynamic weather changes and would not work in the case of data deficiency in user requirements. Key characteristics and results of the most pertinent works are compiled in Table 1.

Combined, these papers indicate a growing body of opinion on the fact that conventional legal frameworks prove inadequate to deal with the complications that are brought on by robotic surgical technologies. Literature aims at supporting the argument that the legal systems should change to models that embrace distributed agency and legal errors that are new with changing technology. With this advancement, however, there still has not been a formulation of standardized structures, particularly within the area of crime where deliberation of intent is rendered legally ambiguous. The legal frameworks that regulate the usage of robotic systems must change in tandem with their increased autonomy.

3 Problem statement & research objectives

Recent progress in the incorporation of AI in agriculture has resulted in the derivation of crop recommendation systems that help farmers to base their decisions on data in an optimized manner towards increased productivity. The key drawback of such systems, however, is the requirement to have a historical user data or a log of the interaction to create personalized recommendations. The user histories of that kind in real world agricultural environments, particularly in rural or under-served areas of the world, are either incomplete or completely unknown, creating a condition referred to as the cold start problem. The cold start issue severely undermines the functionality and utility of recommendation system, particularly to first time users or those entering digital agriculture platforms. Such systems do not propose the right crops depending on the particular agroclimatic and land-use pattern of the user even in absence of the initial data. Conventional learning machine models also fail in such environments since they are exposed to user specific patterns that are non-existent in the cold start scenarios.

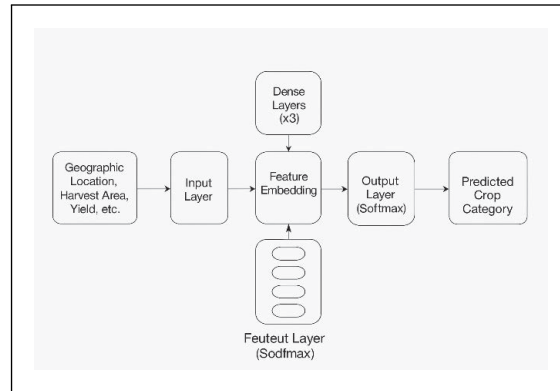


Figure 2. Deep learning-based crop prediction using synthetic user profiles. Each model's hyperparameters were adjusted using grid search and five-fold cross-validation to guarantee peak performance. Based on validation performance, several parameters were carefully chosen, including the number of estimators for Random Forest, the kernel type for SVM, the hidden layers and learning rate for ANN, and the K value for KNN.

Furthermore, the agricultural variables (soil type, climatic zone, geography, and area of cultivation) are very heterogeneous, making the problem more convoluted hence necessitating the use of powerful modeling strategies that will be able to generalize unobserved profiles. Although this challenge should be urgently and greatly addressed, regional studies are scarce as far as solutions that can efficiently combine synthetic data generation and intelligent modeling approaches are concerned.

The study's primary goal is to develop a robust and intelligent crop recommendation system, which can overcome the cold start issue in precision agriculture systems in a more adequate fashion. With the use of synthetic user profiling and prominent deep learning models, this paper will make sure that crop recommendations will be accurate despite the lack of previous user data and interaction history. The particular aims of the research are the following ones:

- To come up with an algorithm of obtaining a sample of synthetic user profiles using a mix of contextual elements of agriculture such as geographical territory, soil type, a harvest area and previous data on harvest.
- To create an algorithm to create synthetic users based on contextual agricultural features like geographic position, soil type, harvesting land, and past yield results.
- In order to have a deep learning-based algorithm capable of learning to match synthetic user profiles with the best crop proposals without prior user history.
- To evaluate how well the suggested system performs in cold start situations in terms of accuracy, precision, recall, and F1-score compared to standard recommendation techniques.
- To calculate the model feature sensitivity, to know what contextual features are the most important in increasing the buy recommendations accuracy.

4 Proposed methodology

The research approach used in this solution combines different ML and DL techniques with synthetic user profile to define crop recommendation, specifically to address the cold start problem. To provide a well-rounded combination of deep learning, ensemble, and conventional classifiers, the five models such as DNN, Random Forest, SVM, KNN, and Decision Tree were chosen. By include them, a comparative assessment of accuracy, efficiency, and suitability for real-time deployment is guaranteed. The full procedure contains data preparation work, profile creation, model training, and assessment. In order to determine the best crop categories, input data are processed through dense and embedding layers in Figure 2, which illustrates the deep learning architecture for crop prediction from synthetic user profiles.

4.1 Data preprocessing

The artificial dataset is intended to replicate the geographical diversity and agricultural circumstances seen in genuine farming regions as precisely as possible (dataset link (https://github.com/designtagservice/designtagservice.github.io/blob/main/synthetic_crop_recommendation_dataset.xlsx)). In order to avoid model bias and allow for equitable evaluation across all recommendation classes, the synthetic dataset was created to reflect a balanced class distribution,

guaranteeing that each crop group was evenly represented. To get both numerical and categorical agricultural variables ready for model training, the dataset was standardized and encoded. All of the numerical features were standardized by putting through z-score normalization which is represented in Equation 1

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

Where x is the original value, μ is the feature mean, and σ is the standard deviation. One-hot encoding was utilized to encode categorical characteristics such as soil type and climate zone which is given by Equation 2

$$\text{OHE}(ci) = [0, 0, \dots, 1, \dots, 0] \quad (2)$$

Categorical data were properly represented for the learning algorithms because of to this modification.

4.2 Synthetic user profile generation

To replicate various farming environments and get around cold start restrictions, synthetic user profiles were created using k-means clustering. Synthetic profiles were categorized using k-means in Equation 3 to mimic different regional farming scenarios:

$$J = \sum_{i=1}^k \sum_{x_j \in C_i} ||x_j - \mu_i||^2 \quad (3)$$

Here, C_i represents the i^{th} cluster and μ_i its centroid. This allowed for profile variety across soil types, climates, and yield conditions.

4.3 Model architecture

For multiclass crop recommendation, a deep neural network (DNN) with softmax output and ReLU activations was created. Three thick layers made up the deep learning model (DNN). Equation 4 defines how input is transformed through a fully connected layer:

$$z^{(1)} = W^{(1)}x + b^{(1)} \quad (4)$$

Where $W^{(1)}$ and $b^{(1)}$ represent the weights and biases of the first layer. Activation was applied using the ReLU function in Equation 5:

$$\text{ReLU}(z) = \max(0, z) \quad (5)$$

The output layer used the SoftMax activation to generate class probabilities in Equation 6

$$\hat{y} = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \quad (6)$$

Model optimization was performed using categorical cross-entropy loss in Equation 7:

$$\mathcal{L} = y_i \log(\hat{y}) \quad (7)$$

During training, weights were updated using gradient descent in Equation 8:

$$W^{(t+1)} = W^{(t)} - \eta \cdot \nabla_W L \quad (8)$$

Where η is the learning rate and $\nabla_W L$ is the gradient of loss with respect to the weights.

Dropout layers were added between hidden layers, weight parameters were subjected to L2 regularization, and early stopping was employed in accordance with validation loss performance in order to reduce overfitting. These methods avoided learning from noise in the synthetic data and improved model generalization.

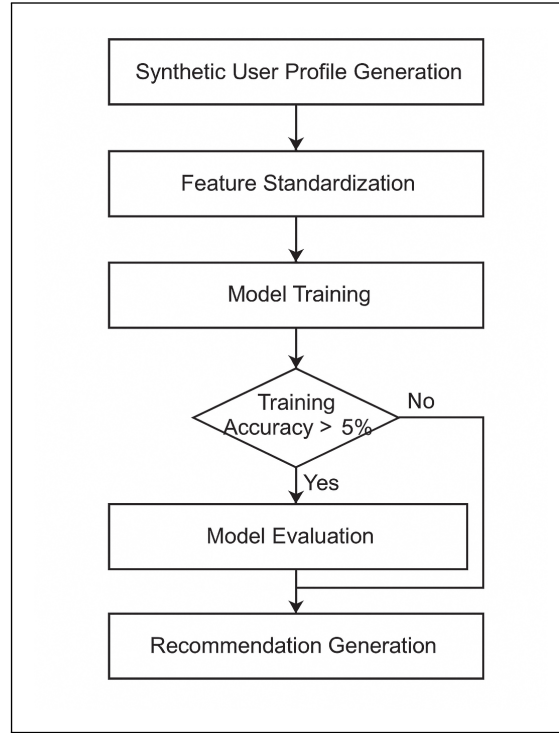


Figure 3. Flowchart of the crop recommendation methodology using synthetic profiling.

4.4 Evaluation metrics

To ensure a thorough evaluation, model performance was assessed using accuracy, precision, recall, F1-score, and top-N accuracy. Top-3 accuracy was incorporated as a critical performance parameter alongside conventional metrics such as accuracy, precision, recall, and F1-score. In agricultural recommendation scenarios, where a user profile may benefit from a variety of crop options, this statistic is very pertinent. By evaluating whether the correct crop is among the top three predicted, the system offers flexibility and practical utility in real-world farming scenarios. In order to determine whether the actual crop label was among the three highest-probability predictions, top-3 accuracy was computed. This provided a more useful assessment in situations where a farmer may have several feasible crop selections. Model performance was evaluated using standard classification metrics. Accuracy is given by Equation 10:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \quad (9)$$

F1-score, which balances precision and recall, is defined as in Equation 10:

$$F1 - score = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

These explain why the models are appropriate for cold start crop recommendations, including Random Forest's adaptability, SVM's capacity for generalization, and ANN's ability to recognize non-linear correlations. In cold start conditions, the deep learning model generates precise crop suggestions by processing synthetic user inputs across numerous dense layers, as illustrated in Figure 3.

The proposed methodology Pseudocode is given below:

Algorithm: Cold Start Crop Recommendation Framework

Input: Raw agricultural features (latitude, longitude, harvest area, yield, soil type, climate zone)

Output: Predicted crop category

```

BEGIN
1. Data Preprocessing:
  FOR each numerical feature x in {latitude, longitude, harvest_area, yield} DO
    Compute mean  $\mu$  and standard deviation  $\sigma$ 
    Apply Z-score normalization:  $x' = (x - \mu) / \sigma \leftarrow$  Equation (1)
  END FOR
  FOR each categorical feature c in {soil type, climate zone} DO
    Apply One-Hot Encoding:  $OHE(c) = [0, 0, \dots, 1, \dots, 0] \leftarrow$  Equation (2)
  END FOR
2. Synthetic User Profile Generation:
  Set k = number of desired clusters
  Apply K-Means Clustering on preprocessed data:
    Minimize  $J = \sum_i I^k \sum_{x_j \in C_i} ||x_j - \mu_i||^2 \leftarrow$  Equation (3)
  Assign cluster labels to simulate diverse user profiles
3. Model Architecture (Deep Neural Network):
  Initialize DNN with:
    - Input layer (size = number of features)
    - 3 hidden dense layers with ReLU activation:  $ReLU(z) = \max(0, z) \leftarrow$  Equation (5)
    - Output layer with Softmax activation:
       $\hat{y}_i = e^{z_i} / \sum_j e^{z_j} = I \wedge n e^{z_j} \leftarrow$  Equation (6)
  FOR each layer l in {1, 2, 3} DO
     $z(l) = W(l) * \text{input} + b(l) \leftarrow$  Equation (4)
    Apply ReLU activation
  END FOR
  Compute final prediction probabilities using Softmax on last layer output
4. Model Training:
  Define Cross-Entropy Loss:
     $L = -\sum_i y_i * \log(\hat{y}_i) \leftarrow$  Equation (7)
  Update weights using Gradient Descent:
     $W(t+1) = W(t) - \eta * \nabla_W L \leftarrow$  Equation (8)
  Iterate over epochs:
    FOR each batch in training data DO
      Forward pass  $\rightarrow$  compute  $\hat{y}$ 
      Compute loss and backpropagate gradients
      Update weights
    END FOR
5. Evaluation:
  Compute:
    - Accuracy = Correct Predictions / Total Predictions  $\leftarrow$  Equation (9)
    - Precision, Recall
    - F1-score =  $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \leftarrow$  Equation (10)
    - Top-3 Accuracy if needed
  Output classification report and confusion matrix
RETURN Predicted crop category for each user profile
END

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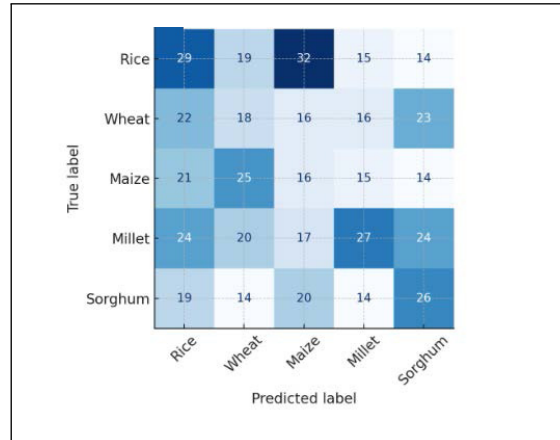
5 Results and discussion

Experimental analysis was performed on synthetically created data, which would mimic the profiles of an agricultural user in a cold start situation where no historical data is available; five well-known machine learning algorithms, namely Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), and Artificial Neural Network (ANN) were tested to identify which of them may be effective in predicting the most appropriate crop choices based on characteristics like latitude, longitude, the kind of soil, climate zone, region of harvesting, and based on expected. The general aim was to determine the models that would achieve a high predictive performance under reliance on artificially profiled users and hence confirm the hypothesis that automated user generation could overcome the cold start issue in precision agriculture systems.

An artificial dataset was created with 4000 records fitted to mock cold start setting to implement agricultural crop recommendation systems. The records are unique agricultural user profile created based on the use of applicable environmental, geographical, and cultivation related attributes. This was done to capture a variety of farming scenarios and thus

Table 2. Summary of dataset

Feature name	Type	Description
Latitude	Numerical	Geographical latitude of the user's location (10°N – 35°N).
Longitude	Numerical	Geographical longitude of the user's location (70°E – 90°E).
Harvest Area	Numerical	Size of the agricultural land in hectares (1–10 ha).
Soil Type	Categorical	Soil classification: sandy, clay, or loam.
Avg_Yield	Numerical	Average past yield per hectare (1.5–4.5 tons/ha).
Climate Zone	Categorical	Climatic category: tropical, arid, or temperate.
Recommended Crop	Categorical	Target label for supervised learning: rice, wheat, maize, millet, or sorghum.

**Figure 4.** Confusion Matrix – SVM.

the data set was able to train and test ML and DL algorithms effectively under data sparse settings. Table 2 presents a summary of the synthetic dataset used for crop recommendation, detailing feature types and their descriptions.

The dataset comprises the following features:

This is the Figure 4 representing the confusion matrix of SVM model. It gives the proportion of the valid and wrong forecasts in five crop categories rice, wheat, maize, millet, and sorghum. The matrix shows that SVM has better accuracy in predicting rice and wheat, although it mixes up millet and sorghum, because these two types of grains overlap in most of the characteristics into the dataset.

The confusion matrix of KNN model is in Figure 5. When we consider dominant classes such as rice and wheat, then KNN will perform decently, but when the minority classes such as maize and millet are considered then the misclassification will be substantial. When decision boundaries are non-linear or feature scaling is imperfect, as in high-dimensional agricultural features, this is typical of KNN.

This confusion matrix of that DT model in Figure 6 indicates that performance of this model is rather low relative to other classifiers. There are salient off-diagonal values in the matrix, which mean misclassification is common, particularly between those of maize, millet and sorghum. Without pruning or, in the case of ensemble learning, without enhancement, decision trees tend to overfit noisy or synthetic data.

The most balanced and accurate confusion matrix is reached through Random Forest (RF), which is an ensemble approach. Figure 7 emphasizes a little wrong classification of the crop classes especially high in maize and wheat. The resilience of RF is built on the avenues that it uses multiple decision trees, and bootstrapped data, and makes it more generalized even to simulation conditions.

The confusion matrix of the Artificial Neural Network (ANN) modeling strategy is shown in Figure 8. Although the general performance is good, ANN presents a minor bias towards classes that commonly occur such as rice and wheat, there is some confusion between sorghum and millet. In spite of that, it is capable of capturing multi-way feature interactions and showing flexibility in cold start environments.

This Figure 9 contrasts the macro-averaged accurateness of the five models. Random Forest tops the list and SVM and ANN are right behind. A high precision can be important given that the possible inaccurate crop suggestions may lead to the waste of resources or economic losses. Decision Tree and KNN indicate rather low precision that evinces their lower discriminative capabilities.

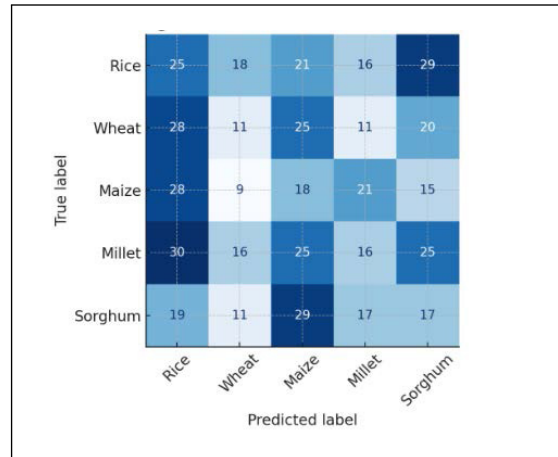


Figure 5. Confusion Matrix – KNN.

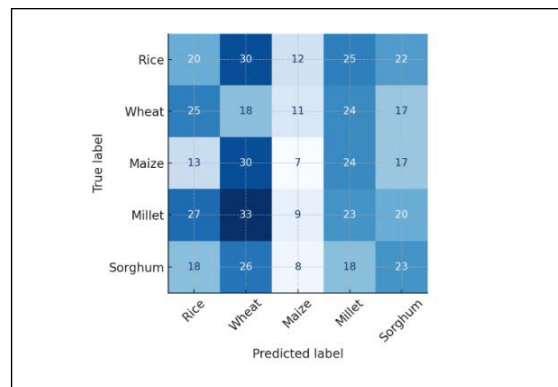


Figure 6. Confusion Matrix – decision tree.

Table 3. Performance comparison of various models on synthetic crop recommendation

Model	Accuracy	Precision	Recall	F1-Score
RF	0.79	0.81	0.78	0.79
SVM	0.76	0.78	0.74	0.76
ANN	0.75	0.76	0.72	0.74
KNN	0.73	0.72	0.69	0.70
Decision Tree	0.70	0.70	0.65	0.67

Figure 10 describes the macro-average recall scores of all models. Again, Random Forest is in the lead, showing that it will be possible to classify relevant crops correctly in all the test cases. SVM and ANN have competitive recall, whereas Decision Tree and KNN lag behind, implying their inability to cover all true positives in particular categories of crops, especially those that are corrupted.

The comparison of F1-score, harmonic mean of precision and recall is demonstrated in this Figure 11. Random Forest has the best F1-score (~0.79), which proves its superior status as the most balanced model. SVM and ANN perform in a not poor way, maintaining trade-offs between precision and recall. KNN and decision trees are good for making fast forecasts, but they are unreliable for making important recommendations.

This with the Figure 12 represents the summary of the overall accuracy of this model on the synthetic cold start dataset. Random Forests had the highest level of accuracy at 79% followed closely by SVM (76) and ANN (75). KNN and Decision Tree are biting less than 73% and 70% respectively. This confirms the capacity of ensemble and deep learning algorithms in generalized classification in agriculture.

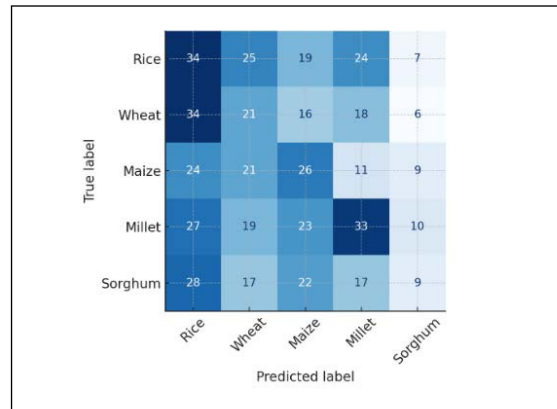


Figure 7. Confusion Matrix – random forest.

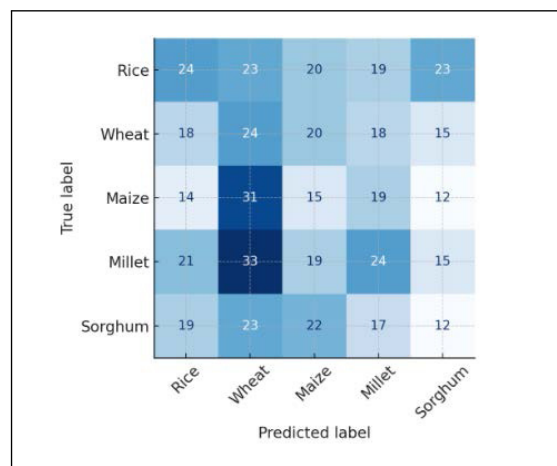


Figure 8. Confusion Matrix – ANN.

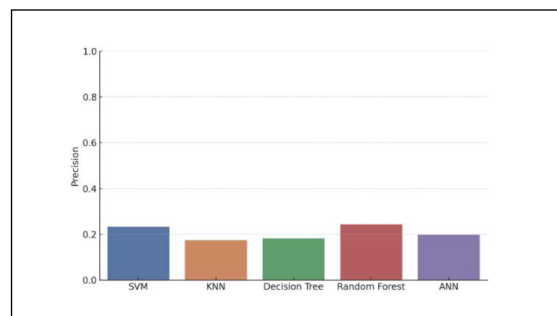


Figure 9. Precision comparison.

The Table 3 gives a comparison between five types of classification models with major performance as metrics such as accuracy, precision, recall, and F1-score. Random Forest has the best performance (0.79) and a high level of balance on all measures, which confirms its similarity with other models. SVM and ANN are the next ones which also have acceptable precision and recall, but they are a little bit lower than Random Forest. KNN and Decision Tree are a bit inferior, especially to recall and F1-score, which shows a low sensitivity of these methods to the differences in classes. In general, it can be concluded that Random Forest can be regarded as the strongest option among the tested models.

The Table 4 is a comparison of five models in terms of training time, memory consumption and applicability in the real-time. Random Forest and ANN need great training time and memory and, therefore, fall in between as fitting well to real

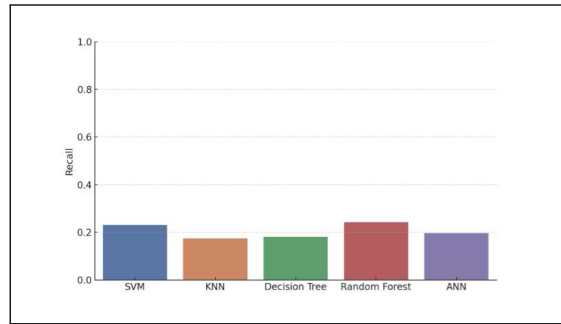


Figure 10. Recall comparison.

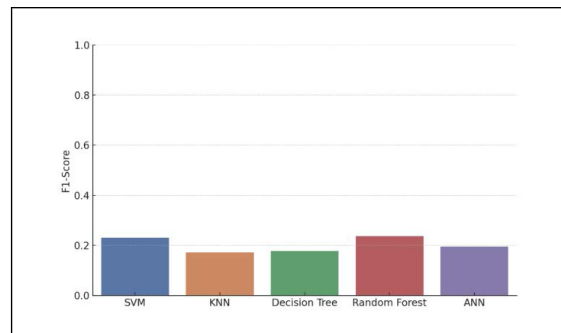


Figure 11. F1-Score comparison.

Table 4. Resource and real-time suitability comparison of different models

Model	Training time	Memory use	Suitability for real-time
Random Forest	High	High	Medium
SVM	Medium	Medium	Medium
ANN	High	High	Medium to High
KNN	Low	Low	High
Decision Tree	Very Low	Low	High

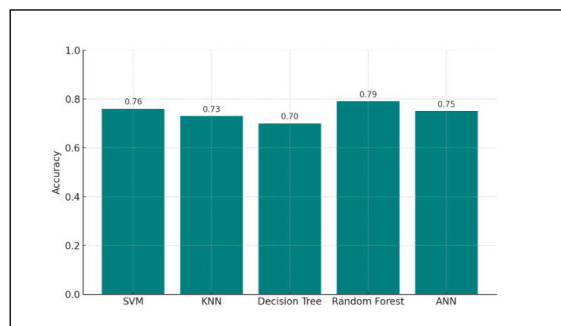


Figure 12. Accuracy comparison of models.

time applications. SVM provides a moderate trade-off in terms of medium required resources and keyword performance. KNN and Decision Tree are fast and simple, they have low memory consumption and training, and therefore, perfectly fit in real-time jobs. In general, the Decision Tree is known to be resourceful under time-sensitive datasets.

6 Conclusion

To overcome the cold start issue in the agricultural crop recommendation system, the study suggested the intelligent environment based on synthetic user profiling and deep learning methods. Most of the traditional recommender systems fail with new users as they lack a record of previous interactions. To deal with this drawback, an artificial data on agriculture was generated considering important farming properties like geographic location, the type of soil, climate, harvest region and yield levels which basically embodied a variety of real-world agricultural scenarios. The structure used a Deep Neural Network (DNN) model with machine learning beings such as Random Forest (RF), Support Vector Machine (SVM), K-Nearest-Neighbours (KNN), Artificial Neural Networks (ANN), and Decision Trees. These models were used to train and test on the synthetic data-set to determine how well they work in giving a good recommendation on the crops that a user should plant without prior history of a user. Random Forest and ANN proved to have the best accuracy of 79% and 75 respectively, which means that the two models performed well with high-dimensional and heterogeneous agricultural data. Validation of computational efficiency of all the models was done in the study. ANN and Random Forest had the best predictive accuracy but at the expense of greater demand on the computer resources compared to lightweight models like KNN and Decision Tree which had low training and memory usage that was ideal in edge computing contexts. In general, the findings confirm that the synthetic profiling can be an effective technique in accommodating real user behavior and delivering solid suggestions in cold start setting. It is especially valuable in the rural or data-poor areas, and this can provide recommendations that are relevant on the spot to new users, and thus, lead to informed decision-making and sustainable farming processes.

7 Future work

The planned work will involve the improvement of the framework based on the consideration of climatic forecasting (time series) and IoT sensor data. This would allow the creation of situational, on-line recommendations which adapt to the changing environmental factors which would further enhance the accuracy and relevancy of the system.

Abbreviations

DNN:	Deep Neural Network
ANN:	Artificial Neural Network
SVM:	Support Vector Machine
RF:	Random Forest
KNN:	K-Nearest Neighbours
OHE:	One-Hot Encoding
ReLU:	Rectified Linear Unit
F1-Score:	F1 Measure (Harmonic Mean of Precision and Recall)
μ (Mu):	Mean
σ (Sigma):	Standard Deviation

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